Blockchain Wallet Setup Guide

📍 Wallet Initialization Location

File: `functions/src/services/blockchain.ts`
Line: 32

Code Flow

```typescript
// 1. Global configuration from environment variables (Lines 3-6)
const BLOCKCHAIN_RPC = process.env.BLOCKCHAIN_RPC || "";
const BLOCKCHAIN_PRIVATE_KEY = process.env.BLOCKCHAIN_PRIVATE_KEY || "";
const CONTRACT_ADDRESS = process.env.CONTRACT_ADDRESS || "";
const BLOCKCHAIN_ENABLED = process.env.BLOCKCHAIN_ENABLED === 'true';

// 2. Global wallet instances (Lines 15-17)
let provider: ethers.providers.JsonRpcProvider | null = null;
let signer: ethers.Wallet | null = null;  // ← Wallet created here
let contract: ethers.Contract | null = null;

// 3. Initialization function (Lines 19-43)
function initializeBlockchain(): void {
  // ...
  provider = new ethers.providers.JsonRpcProvider(BLOCKCHAIN_RPC);
  signer = new ethers.Wallet(BLOCKCHAIN_PRIVATE_KEY, provider);  // ← WALLET
CREATED
  contract = new ethers.Contract(CONTRACT_ADDRESS, CONTRACT_ABI, signer);
  // ...
}
```

🔐 Environment Variables Setup

Step 1: Update `.env` File

Edit `functions/.env` and add blockchain configuration:

```env
# PostgreSQL Configuration (existing)
PG_HOST=34.143.210.224
PG_PORT=5432
PG_USER=postgres
PG_PASSWORD=108bB2212!2025
PG_DB=low_emission_mobility

# Blockchain Configuration (NEW)
BLOCKCHAIN_RPC=https://mainnet.infura.io/v3/YOUR_INFURA_PROJECT_ID
BLOCKCHAIN_PRIVATE_KEY=your_wallet_private_key_here
CONTRACT_ADDRESS=0x...
BLOCKCHAIN_ENABLED=true
```

Step 2: Set Firebase Functions Configuration

Run these commands to set blockchain environment variables:

```bash
cd functions

# Set blockchain RPC
firebase functions:config:set
blockchain.rpc="https://mainnet.infura.io/v3/YOUR_KEY"

# Set private key
firebase functions:config:set blockchain.privatekey="YOUR_PRIVATE_KEY"

# Set contract address
firebase functions:config:set blockchain.contractaddress="0x..."

# Enable blockchain
firebase functions:config:set blockchain.enabled="true"

# Verify configuration
firebase functions:config:get
```

🎯 How Wallet Initialization Works

Flow Diagram

```
1. Deploy Firebase Functions
          ↓
2. initializeBlockchain() is called
          ↓
3. Create JsonRpcProvider (connects to blockchain RPC)
          ↓
4. Create Wallet from BLOCKCHAIN_PRIVATE_KEY
          ↓
5. Create Smart Contract instance with signer
          ↓
6. Log wallet address and contract address
          ↓
7. Ready for blockchain transactions
```

Detailed Code Explanation

```typescript
// Step 1: Load environment variables
const BLOCKCHAIN_PRIVATE_KEY = process.env.BLOCKCHAIN_PRIVATE_KEY || "";

// Step 2: Create provider (RPC connection)
```

```
provider = new ethers.providers.JsonRpcProvider(BLOCKCHAIN_RPC);

// Step 3: Create wallet from private key + provider
// This creates a new ethers.Wallet instance
signer = new ethers.Wallet(BLOCKCHAIN_PRIVATE_KEY, provider);

// The wallet automatically derives:
// - Public key from private key
// - Ethereum address from public key
// - Signer instance for transactions

// Access wallet address
console.log(signer.address);  // e.g., 0x1234...

// Step 4: Create contract with signer for signing transactions
contract = new ethers.Contract(CONTRACT_ADDRESS, CONTRACT_ABI, signer);
```

📝 Wallet Properties

Once initialized, the wallet (`signer`) has these properties:

```typescript
// Wallet Address
signer.address          // e.g., "0x1234...5678"

// Get Balance
await signer.getBalance()   // Returns balance in wei

// Sign Messages
await signer.signMessage(message)

// Sign Transactions
await signer.sendTransaction(tx)
```

🚀 How to Get Wallet Information

Option 1: View Initialized Wallet Address

Edit `functions/src/index.ts` and add:

```typescript
import { blockchainHealthCheck } from "./services/blockchain";

export const getBlockchainStatus = onRequest(async (req: Request, res: Response)
=> {
  try {
    const status = await blockchainHealthCheck();
    res.json(status);
```

```
  } catch (error) {
    res.status(500).json({ error: String(error) });
  }
});
```

Deploy and call:
```bash
curl https://your-project.cloudfunctions.net/getBlockchainStatus
```

Response:
```json
{
  "status": "healthy",
  "details": {
    "network": "mainnet",
    "chainId": 1,
    "blockNumber": 18500000,
    "contractAddress": "0x...",
    "signerBalance": "0.5"
  }
}
```

Option 2: Check Console Logs

After deployment, check Firebase Functions logs:

```bash
firebase functions:log
```

Look for:
```
Blockchain initialized: {
  network: "https://mainnet.infura.io/v3/...",
  contractAddress: "0x...",
  signerAddress: "0x1234...5678"  ← Your wallet address
}
```

🔑 Creating Blockchain Private Key

From MetaMask

1. Open MetaMask → Settings → Security & Privacy
2. Click "Export Private Key"
3. Enter password, copy private key
4. Add to `.env`:
   ```env
   BLOCKCHAIN_PRIVATE_KEY=your_copied_private_key
   ```

   ```

### From ethers.js

```typescript
import { ethers } from "ethers";

// Generate new wallet
const newWallet = ethers.Wallet.createRandom();
console.log("Private Key:", newWallet.privateKey);
console.log("Address:", newWallet.address);
```

📋 Required for Production Deployment

Checklist

- [ ] Have RPC endpoint URL (Infura, Alchemy, Quicknode, etc.)
- [ ] Have wallet private key
- [ ] Have deployed smart contract address
- [ ] Set `BLOCKCHAIN_ENABLED=true`
- [ ] Test wallet has sufficient ETH for gas fees
- [ ] Smart contract has required functions:
  - `redeemVoucher(address user, string voucherId)`
  - `getRedemption(address user, string voucherId)`

Deploy with Blockchain

```bash
cd functions

# 1. Set all environment variables
firebase functions:config:set \
  blockchain.rpc="https://mainnet.infura.io/v3/YOUR_KEY" \
  blockchain.privatekey="YOUR_PRIVATE_KEY" \
  blockchain.contractaddress="0x..." \
  blockchain.enabled="true"

# 2. Deploy functions
firebase deploy --only functions

# 3. Verify
firebase functions:log
```

🔌 Troubleshooting Wallet Issues

Issue: "Blockchain initialization failed"

Solution: Check environment variables
```

```bash
firebase functions:config:get | grep blockchain
```

Make sure all three are set:
- `blockchain.rpc`
- `blockchain.privatekey`
- `blockchain.contractaddress`

Issue: "Invalid Ethereum address"

Solution: Verify private key format
```bash
# Should NOT start with 0x
☑ BLOCKCHAIN_PRIVATE_KEY=abc123def456...
❌ BLOCKCHAIN_PRIVATE_KEY=0xabc123def456...
```

Issue: "Insufficient funds for gas fees"

Solution: Add ETH to wallet
```bash
# Get wallet address
firebase functions:log | grep signerAddress

# Send ETH to that address from another wallet
# Then retry transaction
```

Issue: "Transaction nonce expired"

Solution: Retry the transaction
```bash
# The wallet will use correct nonce on next attempt
# No manual intervention needed
```

📞 Wallet Functions Available

1. recordRedemptionOnChain()

```typescript
const txHash = await recordRedemptionOnChain(
  userAddress,      // User's wallet address
  voucherId         // Voucher ID to redeem
);
```

2. checkRedemptionOnChain()

```typescript
const isRedeemed = await checkRedemptionOnChain(
```

```
  userAddress,      // User's wallet address
  voucherId         // Voucher ID to check
);
```

3. getAccountBalance()

```typescript
const balance = await getAccountBalance();
// Returns balance in ETH (formatted)
```

4. blockchainHealthCheck()

```typescript
const status = await blockchainHealthCheck();
// Returns: { status, details }
```

🔄 Wallet Lifecycle

```
Application Start
       ↓
initializeBlockchain() called (first function using blockchain)
       ↓
Provider created
       ↓
Wallet created from BLOCKCHAIN_PRIVATE_KEY
       ↓
Contract instance created with signer
       ↓
Wallet ready for signing transactions
       ↓
Transaction executed with wallet signature
       ↓
Blockchain records transaction
       ↓
Transaction receipt returned to application
```

📚 References

- **ethers.js Wallet**: https://docs.ethers.org/v5/api/signer/#Wallet
- **Firebase Functions Config**:
https://firebase.google.com/docs/functions/config-env
- **Infura RPC**: https://infura.io/
- **Smart Contract ABI**: https://docs.ethers.org/v5/api/contract/contract-abi/
